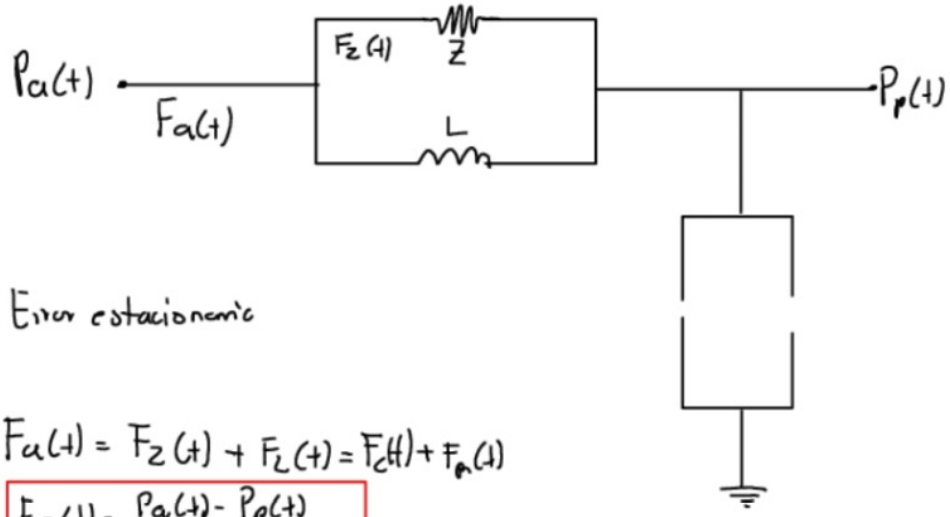
7/Octubre/2026 / Clase Virtual Sistema Cardiovascular



Error estacionario

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_{z(t)} = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int (P_a(t) - P_p(t))\,dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{dt}$$

- Transformada de Laplace

$$F_{z(t)} = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(t) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(t) = C_s P_p(s)$$

$$F_R(t) = \frac{P_p(s)}{R}$$

Relación de nodos con Laplace

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

- Procedimiento algebraico

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$$

$$\left[\frac{1}{Z} + \frac{1}{Ls}\right) P_a(s) = \left[(s + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right] P_p(s)$$

$$\frac{Ls+Z}{LZs}P_a(s) = \frac{CLR2s^2+L(2+R)s+RZ}{LRZs}P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLR2s^2+L(2+R)s+RZ}{LRZs}} = \frac{LRs+RZ}{CLR2s^2+L(2+R)s+RZ}$$

→ Función de Transferencia

- Error estacionario

Este error se calcula cuando $[P_a(s)]$ es un escalon unitario, es decir $[P_a(s)] = \frac{1}{s}$

$$e(s) = \lim_{s=0} sP_a(s) = 1 - \frac{P_p(s)}{P_a(s)} = \lim_{s=0} s\frac{1}{s}\left[1 - \frac{LRs+RZ}{CLR2s^2+L(2+R)s+RZ}\right] = \emptyset$$

$e(t) = 0\ v$ → Error Estacionario

- Estabilidad del sistema en lazo abierto

- Para analizar la estabilidad del lazo abierto se calculan los polos

$\underbrace{CLR2s^2}_{a} + \underbrace{L(2+R)s}_{b} + \underbrace{RZ}_{c}$

$a = CLR2$

$b = L(2+R)$

$c = RZ$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-L(2+R) \pm \sqrt{L^2(2+R)^2 - 4CLR^2Z^2}}{2CLR2}$$

Parámetro	Hipotenso	Normotenso	Hipertenso	Unidades
Z	0.010	0.010	0.010	$mmHg \cdot s \cdot ml^{-1}$
L	0.001	0.001	0.001	$mmHg \cdot s^2 \cdot ml^{-1}$
R	1.000	1.000	1.000	$mmHg \cdot s \cdot ml^{-1}$
C	3.500	1.500	1.000	$ml \cdot (mmHg)^{-1}$

- Hipotenso -

$Z = 0.010$

$L = 0.001$

$R = 1$

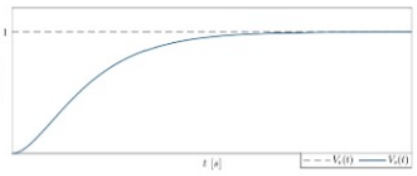
$C = 3.500$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs+RZ}{CLR2s^2+L(2+R)s+RZ} = \frac{(0.001)s+0.01}{(3.5\times10^{-5})s^2+(0.00101)s+0.01}$$

$\lambda_1 = -14.429 + 8.8051i$

$\lambda_2 = -14.429 - 8.8051i$

Por lo tanto, el sistema presenta una respuesta estable y subamortiguada



- Normotenso

$Z = 0.010$
$L = 0.001$
$R = 1$
$C = 1.5$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(2+R)s + RZ} = \frac{0.001s + 0.01}{1.5\times10^{-5}s^2 + 0.00101s + 0.01}$$

$\lambda_1 = -12.062$

$\lambda_2 = -55.272$

Respuesta estable y sobreamortiguada

- Hipertenso

$Z = 0.010$
$L = 0.001$
$R = 1$
$C = 1$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(2+R)s + RZ} = \frac{0.001s + 0.01}{0.00001s^2 + 0.00101s + 0.01}$$

$\lambda_{1,2} = -10$

Respuesta criticamete amortiguada.

- Modelo de ecuaciones integro-diferenciales

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int (P_a(t) - P_p(t))\,dt$$

$$F_C(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{dt}$$

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int (P_a(t) - P_p(t))\,dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{dt}$$

$$P_p(t)\left[\frac{Z+R}{RZ}\right] = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$